

EMORY
UNIVERSITY

DATA521 FINAL PROJECT

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Research Question & Overview

Research Question: How do sector-level S&P 500 stock returns correlate with macroeconomic conditions — including interest rates, inflation, and market volatility?

Bullet points:

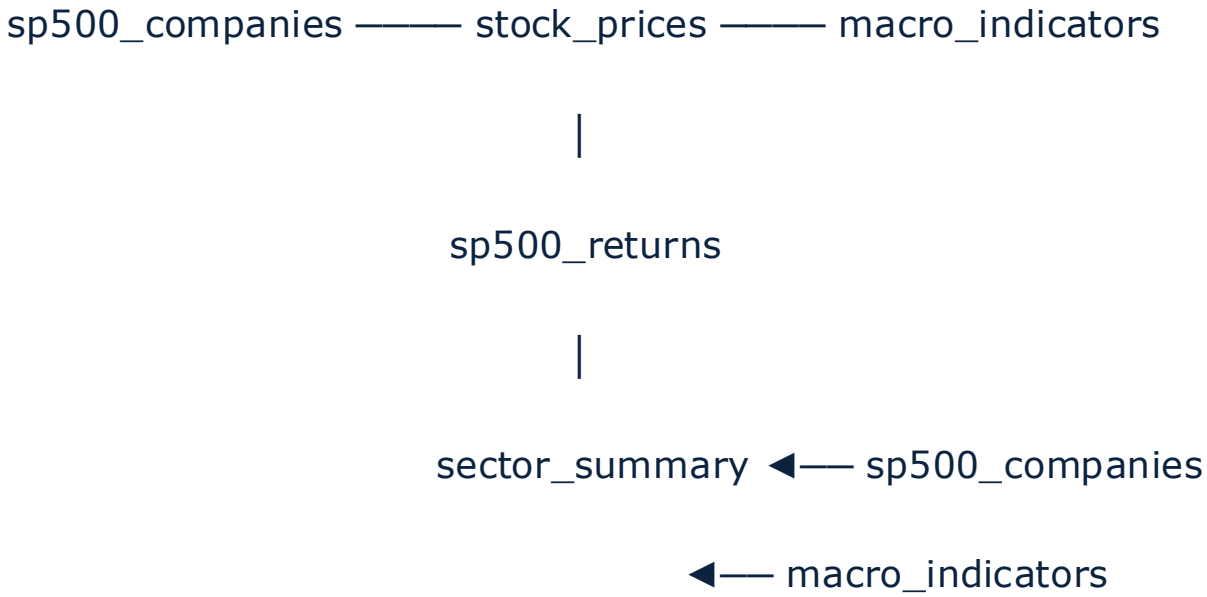
- Time range: January 2013 – February 2018
- Universe: 503 S&P 500 constituent companies across 11 GICS sectors
- Database separates raw source data from derived analytical tables to eliminate redundancy and keep analytical logic transparent
- Two natural join keys across all tables: **ticker** and **date**

Data Sources

Source	Dataset	Description
Kaggle	stock_prices	Daily OHLCV prices per ticker
Wikipedia	sp500_companies	Company metadata for 503 firms
FRED	macro_indicators	Fed Funds Rate, CPT, VIX

FRED data is originally monthly; forward-filled to daily frequency during preprocessing to align with stock price data

Database Architecture



The schema consists of 3 source tables and 2 derived analytical tables, linked by two natural keys: ticker and date.

- "ticker" → joins sp500_companies ↔ stock_prices ↔ sp500_returns
- "date" → joins macro_indicators ↔ stock_prices ↔ sp500_returns ↔ sector_summary

Source Tables

Table 1 — stock_prices (Source: Kaggle)

- One row per ticker per trading day
- Primary Key: price_id (IDENTITY)
- UNIQUE constraint on (ticker, date)
- Columns: open, high, low, close, volume

Table 2 — sp500_companies (Source: Wikipedia)

- One row per company, 503 total
- Primary Key: ticker
- Columns: security, sector, sub_industry, hq_location, date_added, cik, founded

Table 3 — macro_indicators (Source: FRED)

- One row per calendar date, 1,350 total
- Primary Key: date
- Columns: fed_funds_rate, cpi, vix

Derived Tables

Table 4 — sp500_returns

- Derived from: stock_prices
- Stores pre-computed daily return per ticker per day
- Formula: daily_return = (close − open) / open
- Avoids recalculating returns in every query
- UNIQUE constraint on (ticker, date)

Table 5 — sector_summary

- Derived from: sp500_returns + sp500_companies + macro_indicators
- Aggregates data at the sector-day level
- Columns: avg_return, avg_vix, avg_fed_rate, ticker_count
- Directly supports the core research question
- UNIQUE constraint on (sector, summary_date)

Design Decisions

- **Raw vs. derived separation:** Source tables store original data unchanged; derived tables pre-compute returns and aggregates — keeps logic transparent and avoids redundancy
- **Surrogate vs. natural keys:** Source tables use natural PKs (ticker, date); derived tables use IDENTITY surrogate keys for simplicity
- **Forward-fill for macro data:** Monthly FRED series are forward-filled to daily frequency so they can JOIN cleanly with daily stock prices
- **NULL policy:** Core numeric fields in source tables are NOT NULL; optional metadata fields (sector, hq_location, founded) allow NULL
- **UNIQUE constraints** on (ticker, date) prevent duplicate entries across price and return tables

Sample Data Preview

- sp500_companies (first 5 rows):

Symbol	Security	Sector	Sub-Industry	HQ Location
MMM	3M	Industrials	Industrial Conglomerates	Saint Paul, MN
AOS	A. O. Smith	Industrials	Building Products	Milwaukee, WI
ABT	Abbott Laboratories	Health Care	Health Care Equipment	North Chicago, IL
ABBV	AbbVie	Health Care	Biotechnology	North Chicago, IL
ACN	Accenture	Information Technology	IT Consulting	Dublin, Ireland

- macro_indicators (first 5 rows):

Date	Fed Funds Rate	CPI	VIX
2013-01-01	0.14	231.679	18.02
2013-01-02	0.14	231.679	14.68
2013-01-03	0.14	231.679	14.56
2013-01-04	0.14	231.679	13.83
2013-01-07	0.14	231.679	13.79

Data Cleaning Challenges

- **Mixed format in "Founded" column:** Some entries contain formats like "2013 (1888)" — requires string parsing before loading into INT column
- **Monthly → Daily alignment:** FRED macro data is monthly; forward-fill (ffill) applied to propagate values across all calendar days
- **VIX availability:** VIX is only recorded on trading days — gaps exist on weekends and holidays, handled by forward-fill
- **CIK data type inconsistency:** CIK values appear as mixed types in raw data — explicit cast to INT required during ETL
- **Dataset shape confirmed:** sp500_companies: (503, 8) | macro_indicators: (1350, 4)

Summary & Next Steps

Summary:

- Designed a 5-table relational schema to study S&P 500 sector returns vs. macroeconomic indicators
- Clearly separates raw source data from derived analytical tables
- All tables connected via two natural keys: ticker and date
- Preprocessing ensures clean daily-frequency alignment across all datasets

Next Steps (Part 3):

- Load all tables into SQL database
- Run analytical queries to explore sector return patterns
- Visualize correlation between sector returns and Fed rate / CPI / VIX
- Identify which sectors are most sensitive to macroeconomic changes

Thanks